

EC441 MT2 Spring 2025 (Ver 179)

This is closed book, no computer, no calculator exam. The time for the exam is 90 minutes.

Q1. Subnets (25 points)

- (a) If a computer has IP address 116.50.137.187 and netmask 255.192.0.0 what is the network address (use CIDR notation)?
- (b) How many hosts are in a /20 IPv4 network? [Remember that the first and last addresses in a network are reserved as the network itself and the broadcast address.]
- (c) Consider the following IPv4 network: 127.0.0.0/9. You wish to break it into exactly 5 subnets. Design these subnets, specify their CIDR addresses so that they partition the original network (i.e. they do not overlap and collectively they use all addresses of the original network).

Q2 Routing (25 points)

Consider a six-node network (nodes A, B, C, D, E, and F). In this network, costs are bi-directionally equal.

A is connected to B (cost 6) and D (cost 1).

The distance vector of A in this network is

y	$D_A(y)$
A	0
B	4
C	8
D	1
E	2
F	8

- (a) Construct a network graph that is consistent with the information provided.
- (b) For the network you have used in part (a), find the shortest path tree from node F to all other nodes.
- (c) If the cost of the link from A to B increases, what does A do if it is using a Distance Vector routing algorithm?
- (d) If the cost of the link from A to B increases, what does A do if it is using a Link State routing algorithm?

Q3 TCP (25 points)

You are provided the following tshark output (very similar to wireshark, but text based)

Capturing on 'Wi-Fi: en0'

```
1 0.000000 10.239.37.231 → 128.197.173.25 TCP 78 64029 → 20441 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 SACK_PERM
2 0.006724 128.197.173.25 → 10.239.37.231 TCP 78 20441 → 64029 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1250 WS=64 SACK_PERM
3 0.006836 10.239.37.231 → 128.197.173.25 TCP 66 64029 → 20441 [ACK] Seq=1 Ack=1 Win=131200 Len=0
4 0.009929 128.197.173.25 → 10.239.37.231 TCP 66 [TCP Window Update] 20441 → 64029 [ACK] Seq=1 Ack=1 Win=131200 Len=0
5 0.888744 10.239.37.231 → 128.197.173.25 TCP 70 64029 → 20441 [PSH, ACK] Seq=1 Ack=1 Win=131200 Len=4
6 0.896728 128.197.173.25 → 10.239.37.231 TCP 66 20441 → 64029 [ACK] Seq=1 Ack=5 Win=131200 Len=0
7 0.897805 128.197.173.25 → 10.239.37.231 TCP 560 20441 → 64029 [PSH, ACK] Seq=1 Ack=5 Win=131200 Len=494
8 0.897806 128.197.173.25 → 10.239.37.231 TCP 66 20441 → 64029 [FIN, ACK] Seq=495 Ack=5 Win=131200 Len=0
9 0.897911 10.239.37.231 → 128.197.173.25 TCP 66 64029 → 20441 [ACK] Seq=5 Ack=495 Win=130688 Len=0
10 0.897981 10.239.37.231 → 128.197.173.25 TCP 66 64029 → 20441 [ACK] Seq=5 Ack=496 Win=130688 Len=0
11 0.898277 10.239.37.231 → 128.197.173.25 TCP 66 64029 → 20441 [FIN, ACK] Seq=5 Ack=496 Win=131072 Len=0
12 0.901864 128.197.173.25 → 10.239.37.231 TCP 66 20441 → 64029 [ACK] Seq=496 Ack=6 Win=131200 Len=0
```

12 packets captured

Answer the following questions about the TCP connection:

- Is it possible to determine the application layer protocol that was used for this connection? Explain.
- Identify the client and server port number and IP address.
- Identify the packets (by number) that are associated with connection establishment.
- Identify the packets (by number) that are associated with connection termination.
- Which packets (if any) carried message data from the application layer?
- Estimate the round trip time (minimum) for this connection.
- Estimate an upper bound on the transmission rates used by this connection, in Mb/s.

Q4 Error Control (25 points)

Consider a block code specified by the following four codewords:

```
00000
11111
11001
01010
```

- What is the rate of the code?
- What is the minimum distance of the code?
- Determine the possible combinations of error correction and error detection for this block code.
- Design a code with the same (n,k) characteristics that optimizes error detection capability. What is the maximum number of bit errors that can always be detected?